

LAB ASSIGNMENT 3

PROBLEM 1: Write a program to implement a linear regression model on the given dataset and find R2 score.

CODE:

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score

df = pd.read_csv("daily-minimum-temperatures-in-me.csv")
df["Date"] = pd.to_datetime(df["Date"], errors="coerce")
df["Daily minimum temperatures"] = pd.to_numeric(df["Daily minimum
temperatures"], errors="coerce")
df = df.dropna()
df["day_index"] = (df["Date"] - df["Date"].min()).dt.days
X = df[["day_index"]]
y = df["Daily minimum temperatures"]
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.3, random_state=42)
model1 = LinearRegression()
model1.fit(X_train, y_train)
y_pred1 = model1.predict(X_test)
r2_split = r2_score(y_test, y_pred1)
print(f"R2 Score (Train-Test Split): {r2_split}")
model2= LinearRegression()
model2.fit(X, y)
y_pred2 = model2.predict(X)
r2_full = r2_score(y, y_pred2)
print(f"R2 Score (Full Data): {r2_full}")
```

OUTPUT:

```
PS D:\NITJ ACADEMICS\OPENLEARN> python -u "d:\NITJ ACADEMICS\OPENLEARN\LAB_2026\LAB_4_F1.py"
R2 Score (Train-Test Split): 6.481178477035776e-05
R2 Score (Full Data): 0.00010567974538144842
```

PROBLEM 2: Write a program to implement a linear regression model on the given dataset and find the R2 score. (HARDCODE).

CODE:

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split

class MeraLR:
    def __init__(self):
        self.m = None
        self.b = None

    def fit(self, X_train, y_train):
        X_train = np.array(X_train).flatten()
        y_train = np.array(y_train)
        x_mean = np.mean(X_train)
        y_mean = np.mean(y_train)
        num = np.sum((X_train - x_mean) * (y_train - y_mean))
        den = np.sum((X_train - x_mean) ** 2)
        self.m = num / den
        self.b = y_mean - self.m * x_mean

    def predict(self, X_test):
        X_test = np.array(X_test).flatten()
        return self.m * X_test + self.b

    def r2_score(self, y_true, y_pred):
        y_true = np.array(y_true)
        ss_total = np.sum((y_true - np.mean(y_true)) ** 2)
        ss_residual = np.sum((y_true - y_pred) ** 2)
        return 1 - (ss_residual / ss_total)

df = pd.read_csv("daily-minimum-temperatures-in-me.csv")
df["Date"] = pd.to_datetime(df["Date"], errors="coerce")
df["Daily minimum temperatures"] = pd.to_numeric(df["Daily minimum
temperatures"], errors="coerce")
df = df.dropna()
df["day_index"] = (df["Date"] - df["Date"].min()).dt.days

X = df[["day_index"]]
```

```
y = df["Daily minimum temperatures"]
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.3, random_state=42)
model = MeraLR()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
r2 = model.r2_score(y_test, y_pred)

print(f"Slope (m): {model.m}")
print(f"Intercept (b): {model.b}")
print(f"R2 Score: {r2}")
```

OUTPUT:

```
PS D:\NITJ ACADEMICS\OPENLEARN> python -u "d:\NITJ ACADEMICS\OPENLEARN\LAB_2026\LAB_4_F2.py"
Slope (m): 3.436974780922816e-05
Intercept (b): 11.134647281423998
R2 Score: 6.48117847705798e-05
```

PROBLEM 3: Write a program to implement a Multiple linear regression model on the given dataset and find the R2 score.

CODE:

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score
df = pd.read_csv("daily-minimum-temperatures-in-me - Copy.csv")
df.columns = df.columns.str.strip()
df = df.drop(columns=["Unnamed: 2", "Unnamed: 3"], errors="ignore")
df["Date"] = pd.to_datetime(df["Date"], errors="coerce")
df["Daily minimum temperatures"] = pd.to_numeric(df["Daily minimum
temperatures"], errors="coerce")
df["Humidity"] = pd.to_numeric(df["Humidity"], errors="coerce")
df = df.dropna()
df["day_index"] = (df["Date"] - df["Date"].min()).dt.days
X = df[["day_index", "Humidity"]]
y = df["Daily minimum temperatures"]
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.3, random_state=42)
model1 = LinearRegression()
model1.fit(X_train, y_train)
y_pred1 = model1.predict(X_test)
r2_split = r2_score(y_test, y_pred1)
print("R2 Score (Train-Test Split):", r2_split)
model2 = LinearRegression()
model2.fit(X, y)
y_pred2 = model2.predict(X)
r2_full = r2_score(y, y_pred2)
print("R2 Score (Full Data):", r2_full)
```

OUTPUT

```
PS D:\NITJ ACADEMICS\OPENLEARN> python -u "d:\NITJ ACADEMICS\OPENLEARN\LAB_2026\LAB_4_F3.py"
R2 Score (Train-Test Split): 1.0
R2 Score (Full Data): 1.0
_
```

PROBLEM 4: Write a program to implement a Multiple linear regression model on the given dataset and find the R2 score. (HARDCODE)

CODE:

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split

class MeraMultiLR:
    def __init__(self):
        self.w0 = None
        self.w1 = None
        self.w2 = None

    def fit(self, X, y):
        X = np.array(X)
        y = np.array(y)
        ones = np.ones((X.shape[0], 1))
        X_b = np.hstack((ones, X))
        W = np.linalg.inv(X_b.T @ X_b) @ X_b.T @ y
        self.w0 = W[0]
        self.w1 = W[1]
        self.w2 = W[2]

    def predict(self, X):
        X = np.array(X)
        return self.w0 + self.w1 * X[:, 0] + self.w2 * X[:, 1]

    def r2_score(self, y_true, y_pred):
        y_true = np.array(y_true)
        ss_total = np.sum((y_true - np.mean(y_true)) ** 2)
        ss_residual = np.sum((y_true - y_pred) ** 2)
        return 1 - (ss_residual / ss_total)

df = pd.read_csv("daily-minimum-temperatures-in-me - Copy.csv")
df.columns = df.columns.str.strip()
df = df.drop(columns=["Unnamed: 2", "Unnamed: 3"], errors="ignore")

df["Date"] = pd.to_datetime(df["Date"], errors="coerce")
df["Daily minimum temperatures"] = pd.to_numeric(df["Daily minimum
```

```
temperatures"], errors="coerce")
df["Humidity"] = pd.to_numeric(df["Humidity"], errors="coerce")

df = df.dropna()
df["day_index"] = (df["Date"] - df["Date"].min()).dt.days

X = df[["day_index", "Humidity"]]
y = df["Daily minimum temperatures"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = MeraMultiLR()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
r2 = model.r2_score(y_test, y_pred)

print("w0 (Intercept):", model.w0)
print("w1 (day_index):", model.w1)
print("w2 (Humidity):", model.w2)
print("R2 Score:", r2)
```

OUTPUT:

```
PS D:\NITJ ACADEMICS\OPENLEARN> python -u "d:\NITJ ACADEMICS\OPENLEARN\LAB_2026\LAB_4_F4.py"
w0 (Intercept): 5.2735593669694936e-14
w1 (day_index): -1.0191500421363742e-17
w2 (Humidity): 0.9999999999999976
R2 Score: 1.0
```